CCNA Day 4 Study Notes

PING and ARP

Based on the TBN (TungaBadra Networks) Day 4 class handout - Ping and ARP

1. PING

Purpose

- Check network layer (IP) reachability between two devices.
- Check the round trip time (RTT) to reach the destination - and spot packet drops.

The PING utility (application) uses the **ICMP** protocol, which works at the network layer. The sender sends an **ICMP echo request** to the destination IP address, and the destination answers with an **ICMP echo reply**.

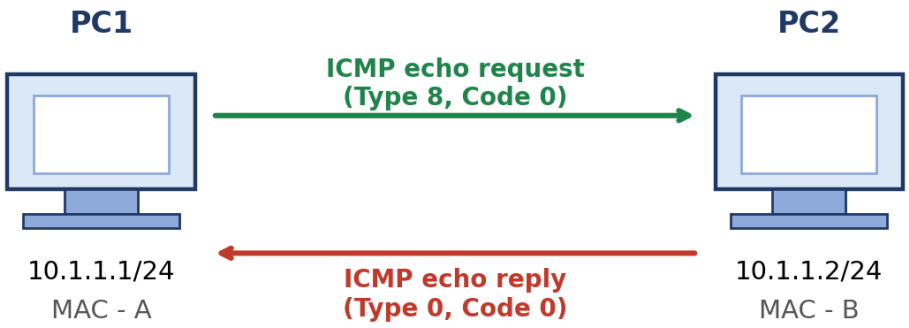
ICMP message	Type	Code	Direction
Echo (ping) request	8	0	Sender -> destination
Echo (ping) reply	0	0	Destination -> sender

How PING works

To check IP reachability from PC1 to PC2: **PC1> ping PC2**

Step 1: PC1 sends an ICMP echo request to PC2.

Step 2: PC2 responds with an ICMP echo reply to PC1's echo request.



In a packet capture (for example Wireshark) on freshly rebooted devices you first see the ARP request/reply pair, and only then the ICMP echo request and echo reply frames, each carrying an identifier, sequence number and TTL. The reply matches the request through the same identifier and sequence number.

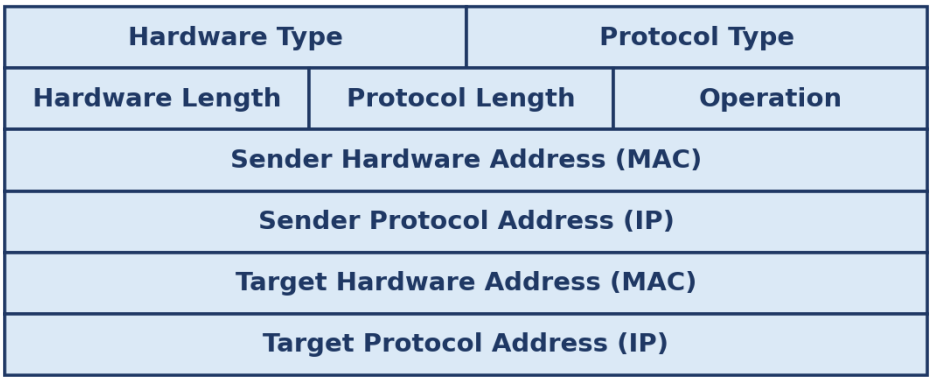
2. ARP - Address Resolution Protocol

ARP features

- ARP is defined in RFC 826.
- ARP is the protocol used by a Layer 3 address (IP) to resolve the Layer 2 address (MAC address) in an Ethernet LAN segment.
- Resolved entries are stored in the ARP cache on Layer 3 devices.
- The sender sends an ARP REQUEST to the Layer 2 broadcast address as the destination.
- The target responds with an ARP REPLY to a Layer 2 unicast address as the destination.

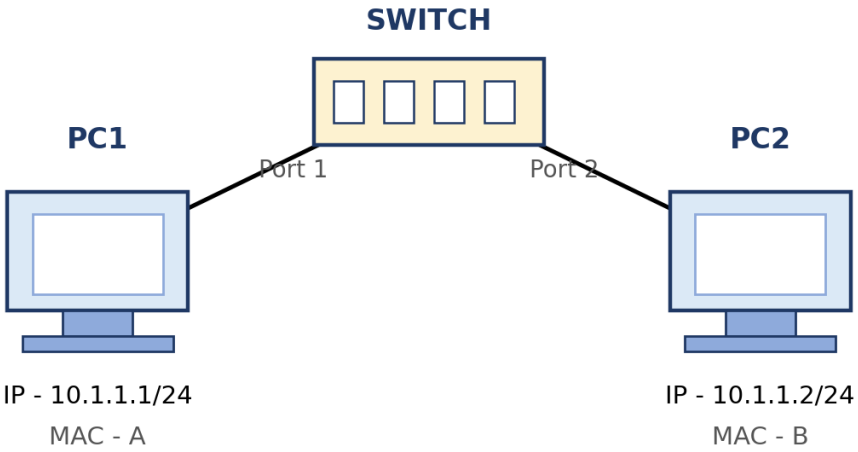
ARP header fields

Field	Size (bits)	Purpose
Hardware Type	16	Type of hardware ARP is running on. Example: Ethernet is type 1.
Protocol Type	16	Higher-level protocol ARP is used for. Example: IPv4 is 0x0800.
Hardware Length	8	Length of the hardware address in bytes. Ethernet MAC = 6 bytes.
Protocol Length	8	Length of the protocol address in bytes. IPv4 = 4 bytes.
Opcode	16	Type of operation: 1 = ARP Request, 2 = ARP Reply, 3 = RARP Request, 4 = RARP Reply.
Sender Hardware Address	48	Physical (MAC) address of the sender.
Sender Protocol Address	32	Logical (IP) address of the sender.
Target Hardware Address	48	Physical (MAC) address of the target.
Target Protocol Address	32	Logical (IP) address of the target.



3. Scenario 1 - ARP in the same subnet

Explain the packet flow to ping PC2 from PC1. (All devices are freshly rebooted.) PC1> ping 10.1.1.2



Step 1: PC1 does an AND operation between the destination IP 10.1.1.2 and its own subnet mask 255.255.255.0 and identifies that it is in the same subnet, 10.1.1.0.

Step 2: Because it is the local (same) subnet, PC1 checks its ARP cache for the MAC address of the destination IP. PC1 was freshly rebooted, so the cache has no entry - it generates an ARP request frame and sends it on the network (Layer 2 broadcast).

Step 3: The switch receives the ARP request frame, checks the I/G bit and understands it is a broadcast frame, so it floods it out of all active ports except the port it arrived on. Before forwarding, the switch learns the source MAC 'A' on port 1 in its MAC address table.

Step 4: PC2 receives the ARP request broadcast frame. In the ARP header it finds its own IP address in the Target IP Address field, and populates its own ARP cache with PC1's IP 10.1.1.1 and MAC 'A'. PC2 generates an ARP reply unicast frame and sends it on the network.

Step 5: The switch receives the ARP reply, checks the I/G bit - this is a unicast frame - looks up the destination MAC 'A' in its MAC address table and forwards the frame out of port 1. Before forwarding, it also learns the source MAC 'B' on port 2.

Step 6: PC1 receives the ARP reply and populates its ARP cache with PC2's IP 10.1.1.2 and MAC 'B'. Now PC1 sends the ICMP echo packet encapsulated in a Layer 2 header onto the network.

Step 7: The switch receives the frame on port 1, checks the I/G bit - unicast - looks up the destination MAC and forwards the frame out of port 2, refreshing the MAC-table entry for 'A'.

Step 8: PC2 receives the frame, sees its own MAC and IP as the destination, and responds with an ICMP echo reply.

Step 9: The switch receives the reply on port 2, checks the I/G bit - unicast - forwards it out of port 1 towards PC1, refreshing the MAC-table entry for 'B'.

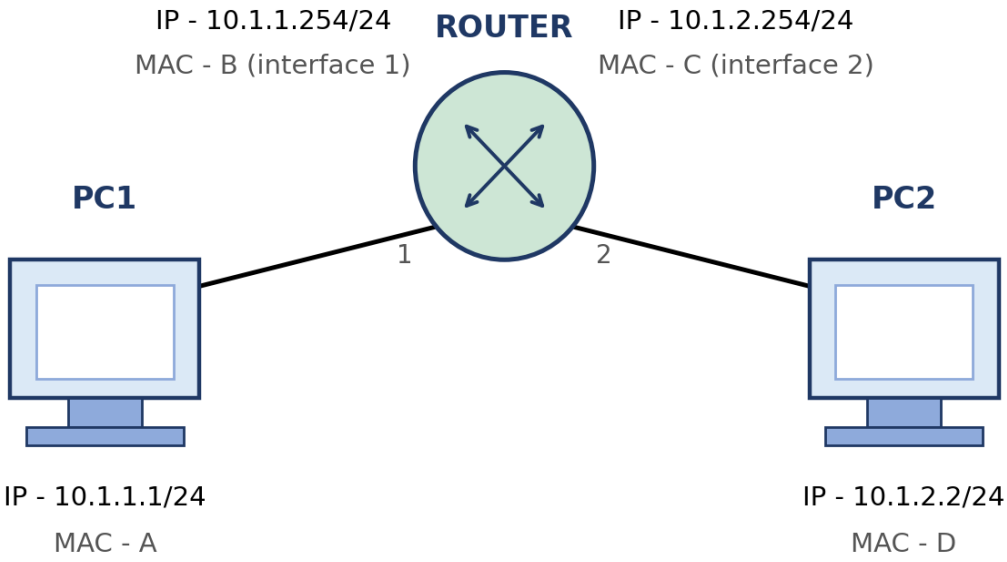
Step 10: PC1 receives the frame, sees its own MAC and IP as the destination, and reads the ICMP echo reply from PC2. Ping succeeds.

Frame contents (ICMP echo request, after ARP resolves)

Layer	Field	Value
L3	Source IP	10.1.1.1
L3	Destination IP	10.1.1.2
L2	SMAC	A
L2	DMAC	? at first -> resolved to B by ARP

4. Scenario 2 - ARP in different subnets

Explain the packet flow to ping PC2 from PC1. (All devices are freshly rebooted.) PC1> ping 10.1.2.2



Step 1: PC1 does an AND operation between the destination IP 10.1.2.2 and its own subnet mask 255.255.255.0 and identifies that 10.1.2.2 is in a different subnet, 10.1.2.0.

Step 2: Because the destination is remote, PC1 does a route lookup in longest-prefix-match order: a) an exact host route (a /32 entry) - no match; b) the longest matching network route - no match; c) the default route (default gateway). The gateway wins, so PC1 needs the MAC of the gateway IP 10.1.1.254. The ARP cache is empty (fresh boot), so PC1 generates an ARP request for the GATEWAY's IP address and broadcasts it.

Step 3: The router receives the ARP request, finds its own IP in the Target IP field, populates its ARP cache with PC1's IP 10.1.1.1 and MAC 'A', and sends an ARP reply unicast frame (its interface MAC 'B').

Step 4: PC1 caches the gateway MAC 'B' and sends the ICMP echo packet: L3 source 10.1.1.1, L3 destination 10.1.2.2, L2 source MAC 'A', L2 destination MAC 'B' (the gateway).

Step 5: The router receives the frame, sees its own MAC as the destination, de-capsulates it and checks the destination IP 10.1.2.2 in its routing table: network 10.1.2.0/24 is directly connected on interface 2. The router's ARP cache has no entry for 10.1.2.2, so it sends an ARP request out of interface 2 and queues the ICMP echo packet meanwhile.

Step 6: PC2 receives the ARP request, populates its ARP cache with the router's IP 10.1.2.254 and MAC 'C', and sends an ARP reply unicast frame (its MAC 'D').

Step 7: The router caches 10.1.2.2 = MAC 'D' and forwards the queued ICMP echo packet towards PC2: the IP addresses are unchanged (10.1.1.1 -> 10.1.2.2 - there is no NAT here; the router still decrements the TTL and recomputes the IP header checksum at each hop, per RFC 791), but it adds a NEW L2 header - SMAC 'C', DMAC 'D'.

Step 8: PC2 receives the frame, sees its own MAC and IP, and sends the ICMP echo reply: L3 source 10.1.2.2, destination 10.1.1.1; L2 SMAC 'D', DMAC 'C'.

Step 9: The router de-capsulates the reply, looks up 10.1.1.1 in its routing table (directly connected on interface 1), finds 10.1.1.1 = MAC 'A' already in its ARP cache, and forwards the reply towards PC1 with a new L2 header - SMAC 'B', DMAC 'A'.

Step 10: PC1 receives the frame, sees its own MAC and IP, and reads the ICMP echo reply from PC2. Ping succeeds.

Frame contents hop by hop

Hop	L3 Source IP	L3 Dest IP	L2 SMAC	L2 DMAC
PC1 -> Router	10.1.1.1	10.1.2.2	A	B
Router -> PC2	10.1.1.1	10.1.2.2	C	D
PC2 -> Router (reply)	10.1.2.2	10.1.1.1	D	C
Router -> PC1 (reply)	10.1.2.2	10.1.1.1	B	A

Key takeaway: Layer 3 (IP) addresses stay the same end to end when no NAT is in use (the router still decrements the TTL and recomputes the IP header checksum at every hop - RFC 791). On Ethernet links the Layer 2 (MAC) addresses are re-written at every hop - the router ARPs for the next hop on an interface when its ARP cache has no entry for it, never for the final destination across a router.

Class handout correction: the class handout lists the route lookup as 'a) Exact host address b) Network Address c) Gateway Address'. The precise mechanism is **longest-prefix match**: the router (or host) picks the route with the longest matching prefix - a /32 host route beats any network route, a more specific network route beats a shorter one, and the default route (0.0.0.0/0, the gateway) matches only when nothing more specific does.

5. Gratuitous ARP (GARP)

Purpose

- Detect a duplicate IP address on the network.
- Refresh tables on the LAN: a GARP is broadcast, so switches learn/refresh the sender's MAC-to-port mapping and other hosts update their ARP caches. (The class handout phrases this as 'educate the switch port about the MAC address'.)

A GARP is sent to the Layer 2 broadcast address and has the **same sender and target IP address** in the ARP header - that equality is its signature.

Section	Field	Value
L2 Header	SMAC	A
L2 Header	DMAC	FFFF.FFFF.FFFF (broadcast)
ARP Header	OpCode	Request or Reply
ARP Header	Sender MAC Address	A
ARP Header	Sender IP Address	10.1.1.1
ARP Header	Target MAC Address	0000.0000.0000
ARP Header	Target IP Address	10.1.1.1 (same as sender IP)

6. Command chart

Every command used in this topic, grouped by device. Commands marked ROOT need administrator / privileged access.

Where	Command	Use
Windows PC	ping 10.1.1.2	Send ICMP echo requests (4 by default) to the target.
Windows PC	ping -t 10.1.1.2	Ping continuously until stopped with Ctrl+C.
Windows PC	ping -n 10 10.1.1.2	Send exactly 10 echo requests.
Windows PC	ping -l 1000 10.1.1.2	Ping with a 1000-byte data payload.
Windows PC	arp -a	Show the ARP cache (IP-to-MAC table).
Windows PC	arp -d *	Clear the ARP cache (ROOT - run as administrator).
Linux PC	ping 10.1.1.2	Ping continuously until stopped with Ctrl+C.
Linux PC	ping -c 4 10.1.1.2	Send exactly 4 echo requests.

Where	Command	Use
Linux PC	ip neigh show	Show the ARP/neighbour cache.
Linux PC	ip neigh flush all	Clear the neighbour cache (ROOT).
Cisco IOS	ping 10.1.2.2	Send 5 ICMP echo requests.
Cisco IOS	ping 10.1.2.2 repeat 100	Send 100 echo requests.
Cisco IOS	show arp / show ip arp	Show the ARP table.
Cisco IOS	show mac address-table	Show the switch MAC address table.
Cisco IOS	clear arp-cache	Clear the ARP cache (privileged mode - ROOT).

7. Interview questions and answers

Q1. What is the purpose of ping?

- Ping checks reachability between two devices - it verifies reachability, latency and packet drops by measuring the round trip time.
- The class handout expands PING as 'Packet InterNet Groper' - that is a well-known backronym, not the original name. Mike Muuss wrote the tool in 1983 and named it after the sound a sonar pulse makes.
- Ping uses ICMP (network layer): an ICMP echo request goes from source to destination (Type 8, Code 0) and the destination answers with an ICMP echo reply (Type 0, Code 0).

Q2. What is the full form of ICMP and its protocol number?

- ICMP = Internet Control Message Protocol. Its IP protocol number is 1.

Q3. How many bytes do Windows and Linux send when we ping?

- Windows sends a 32-byte data payload by default.
- Linux sends 56 bytes of data plus the 8-byte ICMP header, i.e. 64 bytes of ICMP packet - the class handout's 'Windows 32 bytes / Linux 64 bytes'.

Q4. What is the purpose of ARP?

- ARP (Address Resolution Protocol) finds the MAC address for a given destination/next-hop IP address, and stores that MAC in the ARP table (cache).

Q5. Explain the purpose of ARP tables in networking.

- ARP tables keep the mapping of IP addresses to MAC addresses on a local network segment.
- Devices use the table to resolve IP to MAC without broadcasting an ARP request for every single packet.

Q6. ARP is which layer protocol?

- Class answer: Layer 2 protocol.
- Nuance for interviews: ARP carries Layer 3 (IP) addresses inside Layer 2 frames, so many books describe it as operating between Layer 2 and Layer 3 (a Layer 3 helper). If asked, give the class answer and this clarification.

Q7. Who generates the ARP request?

- The PC (host) that needs to resolve a MAC address generates the ARP request.
- If the destination is remote, the host ARPs for its default gateway's MAC instead.

Q8. What is Gratuitous ARP? How does it check the network?

- A GARP is an ARP broadcast where the sender IP and target IP are the same.
- It is used to detect duplicate IP addresses on the network, and it lets switches and hosts refresh their MAC/ARP tables with the sender's address.

Q9. What is Proxy ARP?

- A router answers ARP requests on behalf of another device (replying with its own MAC), so hosts can reach devices on another subnet without a configured gateway.

Q10. What is Reverse ARP (RARP)?

- A device that knows only its own MAC address asks a server to learn its IP address.

Q11. What is the difference between 'Request timed out' and 'Destination host unreachable'? (added by Instinct - beyond the class handout)

- Request timed out: the echo request went out but no reply came back (target down, firewall dropping ICMP, or no return path).
- Destination host unreachable: a router (or the PC itself) could not find a path/next hop and sent back an ICMP error - usually a missing route or gateway problem.

Q12. What is ARP spoofing (ARP poisoning)? (added by Instinct - beyond the class handout)

- An attacker sends fake ARP replies so victims map the attacker's MAC to another host's or the gateway's IP - enabling man-in-the-middle attacks.
- Mitigations: static ARP entries for critical hosts and Dynamic ARP Inspection (DAI) on switches.

Q13. What happens to Layer 2 and Layer 3 addresses when a packet crosses a router? (added by Instinct - beyond the class handout)

- With no NAT in use, the L3 (IP) source and destination addresses stay the same end to end - the router still decrements the TTL and recomputes the IP header checksum at each hop (RFC 791).
- On Ethernet links the L2 (MAC) addresses are re-written at every hop - each router ARPs for the next hop on an interface when its ARP cache has no entry for it.

Q14. Do ARP cache entries stay forever? (added by Instinct - beyond the class handout)

- No - dynamic entries age out. For example, Cisco devices default to a 4-hour ARP timeout and Windows ages dynamic entries much faster. Aging keeps the table accurate as devices change.

8. Quick revision sheet

- Ping = ICMP echo request (Type 8, Code 0) out, echo reply (Type 0, Code 0) back. ICMP is IP protocol number 1.
- ARP (RFC 826) resolves IP -> MAC on a LAN. Request = L2 broadcast, Reply = L2 unicast. Results sit in the ARP cache.
- Same subnet: host ARPs for the DESTINATION's MAC directly; the switch floods the request and learns MACs per port.

- Different subnet: host does a longest-prefix route lookup (host -> network -> default gateway) and ARPs for the GATEWAY's MAC.
- Router behaviour: de-capsulates, route lookup, ARPs on the exit interface on a cache miss (Ethernet links), re-writes L2. IP addresses stay unchanged end to end unless NAT is in use; TTL and header checksum still change at every hop.
- GARP: sender IP = target IP, broadcast - detects duplicate IPs and refreshes tables.
- Windows ping payload 32 bytes; Linux 64 bytes total (56 data + 8 ICMP header).
- Show ARP: arp -a (Windows/Linux), ip neigh show (Linux), show arp (Cisco).